

AMAN JAIN

Pune, Maharashtra

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EDUCATION

Pimpri Chinchwad College Of Engineering

B.Tech - Information Technology

2022 – 2026

Pune, Maharashtra

TECHNICAL SKILLS

Languages: Python, C++, JavaScript

Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph

Databases: MongoDB, MySQL

Developer Tools: VS Code, Git, GitHub, IntelliJ IDEA, Jupyter Notebook, Google Colab

EXPERIENCE

The Idea Company 

Mar 2025 – Jun 2025

AI Developer

- Engineered a startup idea generation platform leveraging **Retrieval-Augmented Generation (RAG)** with a locally hosted **GPT4All-Falcon** LLM, integrating semantic search using **FAISS** and **TF-IDF** over a curated dataset of **500+ cross-domain startup concepts** to deliver context-aware recommendations.
- Designed and implemented backend APIs using **FastAPI** and developed an interactive interface with **Streamlit**, enabling seamless **MERN stack** integration and optimizing end-to-end idea discovery workflows.

PROJECTS

Medical Domain Small Language Model  | [Python](#), [PyTorch](#), [CUDA](#), [Tokenizers](#), [NumPy](#)

- Built a **50.55M-parameter decoder-only Transformer** from scratch (**10 layers, 8 heads, 512 dim**) using **SwiGLU activations**.
- Curated a **764.7M-token biomedical corpus** and trained a custom **Byte-Level BPE tokenizer (16k vocab)** for clinical text.
- Optimized pre-training for **31,000 iterations** on a single **NVIDIA P100 GPU** utilizing **FP16 mixed precision** and **gradient clipping**.
- Eliminated data leakage via **memory-mapped shuffling**, achieving a final validation loss of **3.66 (Perplexity ~38.9)**.

RAG-Based Nutrition Question Answering System  | [Python](#), [PyMuPDF](#), [Hugging Face](#)

- Designed and implemented a **custom Retrieval-Augmented Generation (RAG)** pipeline from scratch for accurate document-grounded question answering.
- Extracted and structured **500+ pages** of PDF data using **PyMuPDF**, generating **1,500+ semantically meaningful chunks** with optimized sliding-window chunking.
- Embedded text chunks using **all-MiniLM-L6-v2** models and implemented cosine similarity search for efficient retrieval.
- Integrated a **quantized Gemma-2B-IT** open-source LLM for GPU-efficient answer generation, providing answers with **page-level citations and relevance scores**.

Machine Learning-Based Exercise Classification System  | [Python](#), [Scikit-Learn](#), [TensorFlow](#)

- Developed a machine learning system to classify barbell exercises in real-time using smartwatch **accelerometer** and **gyroscope** sensor data.
- Collected and processed time-series data for 5 exercises: **squat, bench press, overhead press, barbell row, and deadlift**.
- Engineered features using **FFT, PCA, and K-Means Clustering** to improve model discrimination across movement patterns.
- Trained and benchmarked **KNN, Random Forest, and Neural Network** models, achieving **98.51% classification accuracy**.

COURSEWORK / SKILLS

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|--------------------------------|---------------------|-------------------------------------|---------------------|
| • Data Structures & Algorithms | • Operating Systems | • Database Management System (DBMS) | • Computer Networks |
| | • Machine Learning | | • OOPS Concept |

ACHIEVEMENTS

- Achieved the **Knight badge** in LeetCode contests (**Max. Rating: 1848**).
- Deployed multiple **Transformer-based models** on **Hugging Face Hub**, achieving **300+ community downloads** and external usage.
- Ranked among the **top 10** out of over 70 teams in Datathon, a Data Science Hackathon.